

Robotics & Automation Tasks & Instructions — CodeAlpha

Robotics & Automation — Internship Overview

This internship program provides practical training in robotics design, automation systems and control technologies. CodeAlpha is committed to preparing students for the future of intelligent machines and industrial automation. The internship equips participants with skills in robot kinematics, programming, sensor integration, and process automation. Interns will gain exposure to robotic simulations, industrial control systems and real-world applications in manufacturing, healthcare and smart industries. Guided mentorship and project-based tasks will help interns build a strong foundation to contribute effectively to the field of robotics and automation.

Internship Perks

- Internship Offer Letter
 - Completion Certificate (QR Verified)
 - Unique ID Certificate
 - Letter of Recommendation (based on performance)
 - Job Opportunities / Placement Support
 - Resume Building Support
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Instructions for Interns

1. Share your internship status on **LinkedIn**, tagging **@CodeAlpha**.
 2. Complete the **assigned projects** within the mentioned time frame.
 3. Upload your complete file to **Google drive**:
`CodeAlpha_ProjectName`
 4. Post a **video explanation** of your project on LinkedIn.
 5. Submit your completed task using the **Submission Form**.
 6. **Complete any 2 or 3 out of the 4 tasks** listed below (from your domain).
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Robotics & Automation Task List

(Complete any 2 or 3 of the following tasks)

TASK 1: Research Report on Robotics Applications

Prepare a **detailed report (1500–2000 words)** on robotics in one of the following areas:

- Healthcare (surgical robots, rehabilitation)
 - Manufacturing & Industry 4.0
 - Autonomous vehicles & drones
 - Service robots (domestic, military, agricultural)
- 👉 Submit in PDF with proper references.
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✓ TASK 2: Robotic Arm Simulation

Design a **2D/3D model of a robotic arm** using AutoCAD/SolidWorks/any simulation software.

- Define degrees of freedom (DOF).
 - Perform a basic **motion simulation** (pick-and-place).
 - 👉 Submit CAD/simulation files + screenshots.
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✓ TASK 3: Automation System Design

Create a **flowchart or circuit diagram** for an automated system such as:

- Smart home lighting system
 - Conveyor belt with sensors
 - Automated packaging system
 - 👉 Submit design file + short explanation report (2–3 pages).
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✓ TASK 4: Mini Project – Robotics/Automation Proposal

Prepare a **mini project (5–6 pages)** on one of the following:

- IoT in robotics and automation
 - Role of AI in autonomous robots
 - PLC-based automation system design
 - Humanoid robot design concepts
 - 👉 Submit project report in PDF format.
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Important Note

Internship Completion Criteria:

To be eligible for the internship certificate, participants must complete a minimum of **two or three tasks**. Submitting only one task will be considered **incomplete** and certificates will **not** be issued in such cases.



Submission Details

A submission form will be shared in your respective **WhatsApp group**. You are required to submit your completed task only through that form. Please follow the instructions mentioned in the form carefully to ensure your submission is accepted.



Contact Information

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